

5 Algorithm and Logic

The line follower, in general can encounter three scenarios:

1. Straight path (line) ahead
2. Right turn ahead
3. Left turn ahead

Now, there is no problem when the path ahead is straight. The robot will keep on moving ahead until it encounters any bend or turn. When it encounters a turn, it has to change its direction of locomotion. When a turn towards left arises, it has to turn left and when a right turn is encountered it has to turn right. The direction of turn is recognised by sensors in front. When a left turn arises, the sensor S1 senses the colour of the surface beneath it changing from white to black while S2 is still white surface below it. This condition of sensors is different from the condition when the robot is on straight path and both of its sensors, S1 and S2 are sensing white. Similarly, when a right turn is encountered, the S2 senses a black surface below it instead of white, while S1 continues to sense white surface. This helps the processor conclude that there is a right turn ahead and processor takes necessary steps. These conditions are tabulated below:

S1 (Left sensor)	S2 (Right Sensor)	Conclusion
White	White	Straight path ahead
Black	White	Left turn ahead
White	Black	Right turn ahead

The fourth condition, of both sensors sensing black, won't arise if all the angles in the path are acute. Including obtuse angles and really sharp turns in the path will ask for more efficient algorithm and code, and can be ignored for the time being.

One of the most important things to consider here is the positioning of the sensors. Since, we are following the detection scheme in which the white colour detection would mean the 'normal' condition and black colour detection would mean that there is a turn ahead, so we need to mount out sensors in such a way that they lie just outside the thickness of the black line. The correct way to mount the sensors will be discussed in the chapter 'Assembling and making it work'.

At this stage we have studied about the various options available to us in